Analisis de la temperatura en la Región Ica en GOOGLE

Estadistica

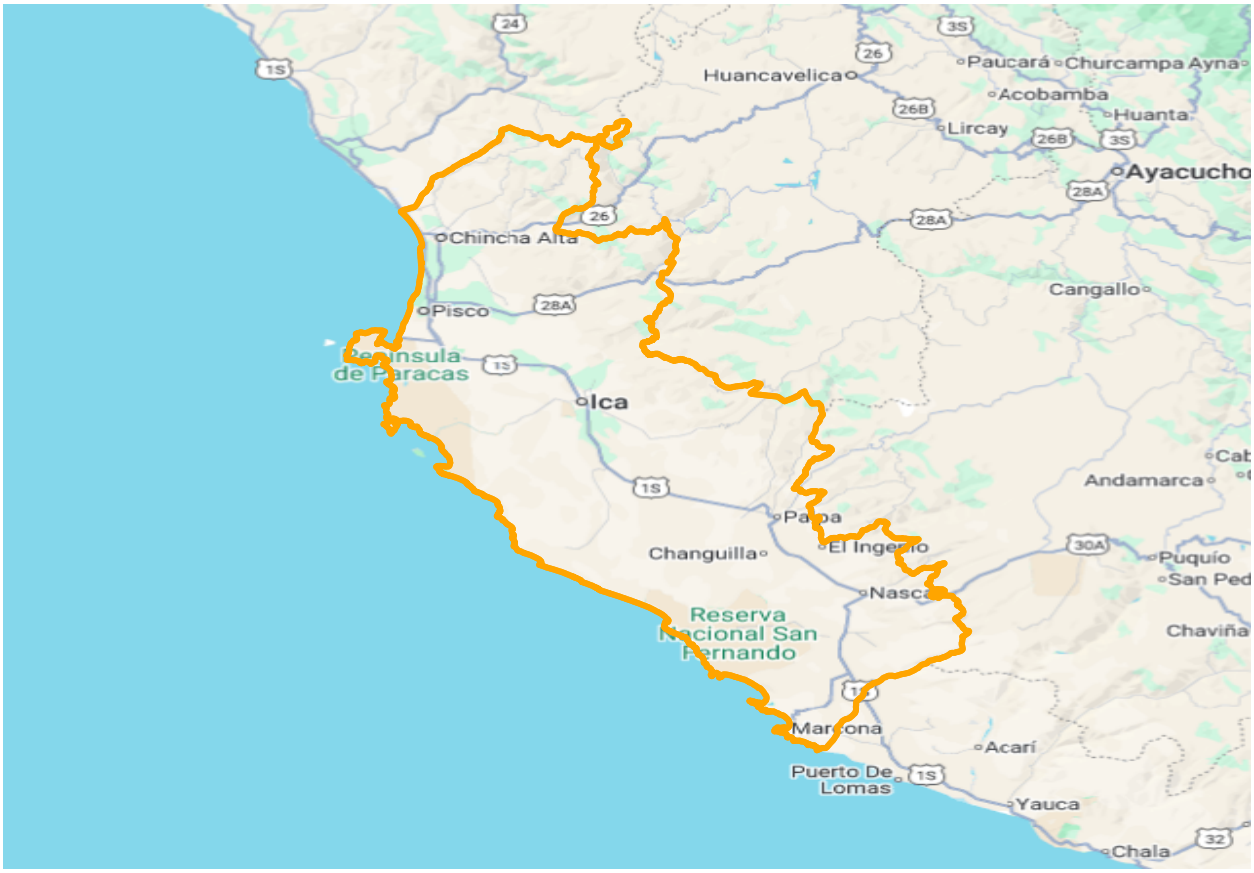
2024-05-21

Region ICA

```
setwd("~/Documentos/Estadistica espacial/Mapas departamentos/Ica")
region<- read.table("Region Ica.txt",header=T)
AMT<-read.table("temperatura ICA.txt",header=T)
peru <- st_read("DEPARTAMENTOS.shp")[0]
```

```
Reading layer `DEPARTAMENTOS' from data source
  `/home/felipe/Documentos/Estadistica espacial/Mapas departamentos/Ica/DEPARTAMENTOS.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 25 features and 79 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -81.32823 ymin: -18.35093 xmax: -68.65228 ymax: -0.03860597
CRS:           NA
```

```
ica<-peru[11,1]
# Extraer las coordenadas del borde de Ica
xy<- ica$geometry
s <- xy[[1]][[1]][[1]]
coords <- s[,c(2,1)]
center <- c(mean(coords[,1]), mean(coords[,2]))
zoom <-8
MyMap <- GetMap(center=center, zoom=zoom, maptype = "roadmap")
LatLon<-LatLon2XY.centered(MyMap, coords[,1],coords[,2],zoom=zoom)
puntos<-cbind(LatLon[[1]],LatLon[[2]])
PlotOnStaticMap(MyMap)
polygon(puntos,border="orange",lwd=3)
e<-c(range(puntos[,2]),range(puntos[,1]))
#plot(grd,add=TRUE)
xy <- LatLon2XY.centered(MyMap, region[,3],region[,4],zoom=zoom)
mes<-AMT[AMT$mes==11,]
tmedia<-apply(mes[,3:4],1,mean)
Tmes <- data.frame(region,temp=tmedia)
Tmes$latitud <- xy$newX
Tmes$longitud<- xy$newY
polygon(puntos,border="orange",lwd=3)
```



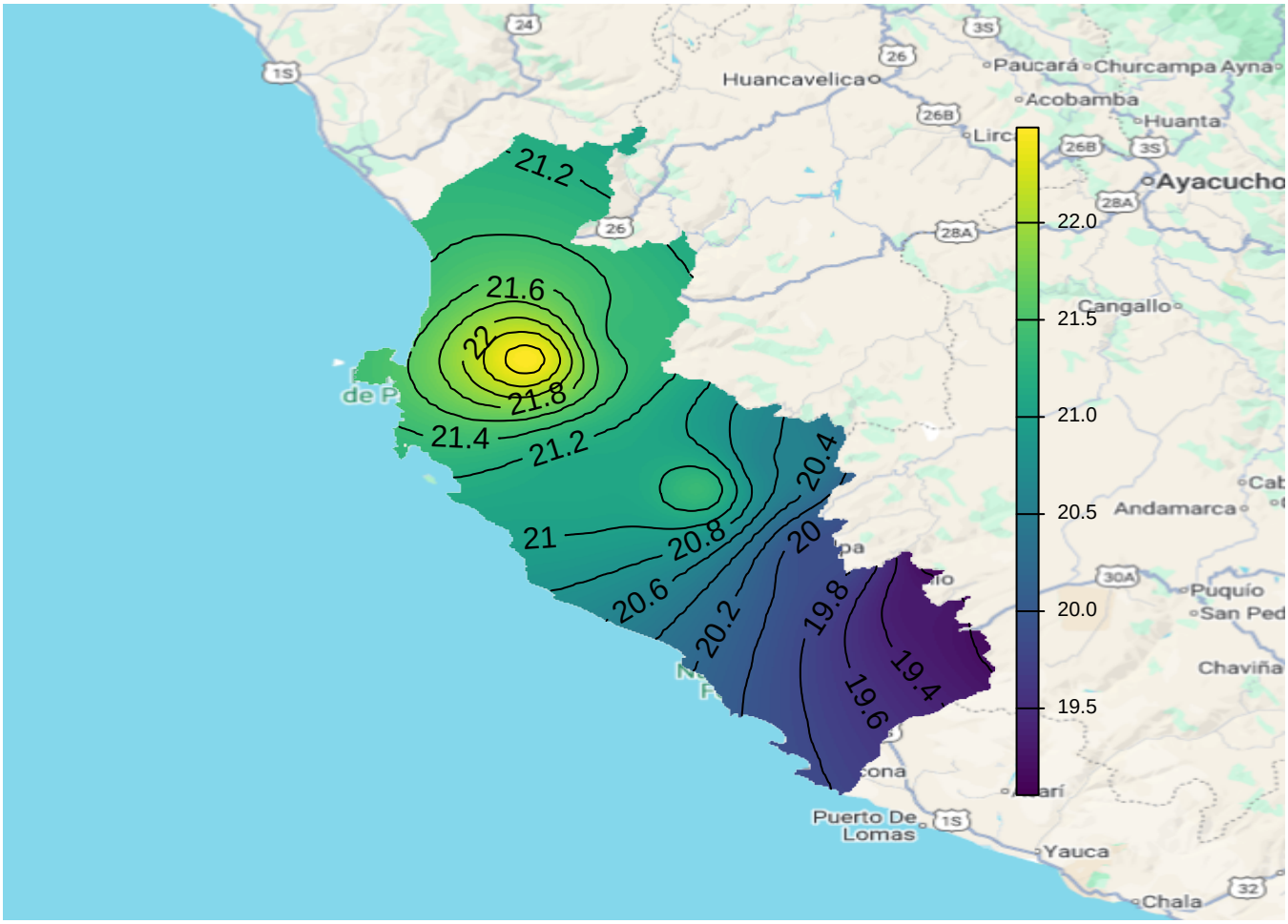

```
# AREA para interpolar
#e<-ext(ica)
grd<-expand.grid(y=seq(from=e[1],to=e[2],by=1), x=seq(from=e[3],to=e[4],by=1)) # by es la distancia
coordinates(grd)<- ~x + y
gridded(grd)<- TRUE
```

Region Ica y mapa de temperatura media para un mes

```
op<-par(mar=c(3,3,3,0),cex=0.8)
coordinates(Tmes)<- ~longitud + latitud
mapa1<-idw(temp ~ 1, Tmes,grd)
```

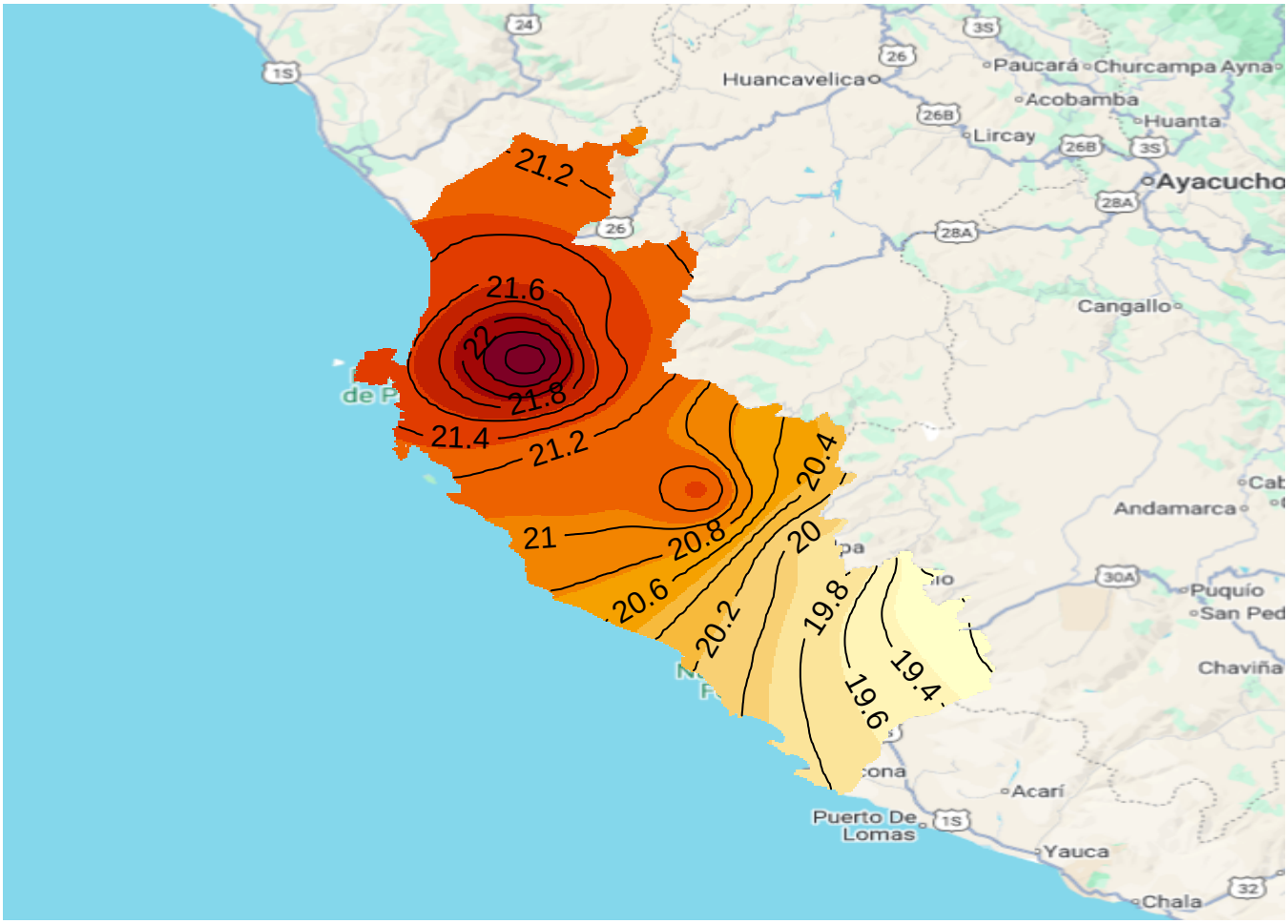
```
[inverse distance weighted interpolation]
```

```
mapa1<-mapa1[,-2]
mapa1<-rast(mapa1)
#.............
pol = st_sfc(st_polygon(list(cbind(puntos[,1],puntos[,2]))))
h<-st_sf(r=10,pol)
ica1 <- mask(mapa1,h)
#...................
PlotOnStaticMap(MyMap)
plot(ica1,add=TRUE)
contour(ica1,add=TRUE, labcex=1,nlevels=20)
```




```
par(op)
```

```
op<-par(mar=c(3,3,3,0),cex=0.8)
#...................
PlotOnStaticMap(MyMap)
image(ica1,add=TRUE)
contour(ica1,add=TRUE, labcex=1,nlevels=20)
```


21.2
21.6
22
21.8
21.4
21.2
21
20.8
20.6
20.4
20
20.2
19.8
19.6
19.4
Huancavelica
Paucará
Churcampa
Ayna
Acobamba
Huanta
Lircay
Ayacucho
Cangallo
Andamarca
Puquío
Chaviña
Acarí
Puerto De Lomas
Yauca
Chala

```
par(op)
```